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A DERIVATION IN EIGHT PARTS

Gradient Residuals

The Calculus of Clearance

Research Note III: The Informational Ground of Nothing

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Gradientology

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CORE CONSTANTS

`log2(3) = 1.585 bits (forbidden ceiling) · log2(10) = 3.322 bits (axis address) · H_B = 0.971 bits`

Abstract

Essay III derives the informational architecture of nothing's structural space from the lattice of Essays I and II. The framework used throughout is pure counting: how many distinct positions exist in a given region of the lattice, and how many bits of binary address are required to distinguish them. No external communication theory, no probabilistic noise model, and no channel-capacity formula is imported. Every result follows from integer arithmetic on the $n = 10$ discrete lattice and the logarithm function — both of which are structural properties of nothing's lattice, not external impositions.

Part I establishes the addressing arithmetic: $n = 10$ positions per axis require $\log_2(10) \approx 3.32$ bits per axis; the full lattice requires $\lceil 3 \times \log_2(10) \rceil = 10$ bits to address any position. Part II derives the face-entropy: the maximum entropy of a three-face label is $\log_2(3) \approx 1.585$ bits, achieved only in the degenerate uniform partition that T.NE (Essay I) forbids. Part III derives the binary axis-entropies of each axis from its clearance fraction, and establishes the axis-entropy ordering $H_B > H_R > H_S$ — which matches exactly the clearance ordering $\sigma > \Lambda > I_{min}$. Part IV derives each clearance margin's addressing bit-count: *Source margin = 1 bit* (exactly), *Boundary margin = 2 bits* (exactly), *Remainder margin* $= \log_2(2.984) \approx 1.577$ *bits* (non-integer, sub-lattice). The one-bit step from Source to Boundary is the informational expression of the Unique Combinatorial Lock's $c/a = 2$ *ratio*. Part V derives the *TSV*'s informational character: its $336/1000$ lattice density gives an information content of $\log_2(125/42) \approx 1.574$ *bits*, which falls precisely one structural bit below $\log_2(3)$ — the informational expression of *TI* exceeding $1/3$ by exactly $1/375$. Part VI derives the dimensional collapse as information compression: $1/d = 1/3$ of the full coordinate information is preserved. Part VII derives the Structural Pixel's information content and the Ontological Shadow's role as the minimum-information 2D clearance cell. Part VIII states the Grand Informational Partition.

Keywords

Addressing Bits · **Face-Entropy** · **Forbidden Ceiling** · **Axis-Entropy Hierarchy** · **Binary Clearance Address** · **Sub-Lattice Information** · **Information Compression** · **Pixel Information** · **Minimum Binary Distinction** · **Grand Informational Partition**

Introduction

The Informational Register

Essays I and II established nothing's structure in two registers: the logical (the Triad, its three faces, the clearances derived from their failure modes) and the geometric (nothing's three-dimensional structural space $NS = [0,1]^3$, its discrete lattice, and the explicit positions of the three clearance margins, the Triadic Sub-Volume, and the Structural Pixel). Essay III reads the same structure in a third register: the informational.

The informational register asks a precise and limited question: given the lattice of Essay II, how many distinct positions exist in each structural region, and how many bits of binary address are required to distinguish them? This question has an exact answer derivable from counting alone. The answer — expressed in bits — is the informational content of each structural region. Nothing more is imported.

This is not Shannon's communication theory. Shannon's theory concerns the capacity of a noisy channel to transmit a message — a theory that requires a probabilistic noise model, a sender, a receiver, and a time-ordered sequence of symbols. None of these are present in nothing's static structural space. What IS present is a discrete lattice of positions. Counting those positions and taking their logarithm base 2 is pure combinatorics — as internal to the lattice's structure as computing the volume of the Triadic Sub-Volume.

The Boundary Condition

One specific formula is explicitly excluded from Essay III: Shannon-Hartley's channel capacity $C = log2(1 + SNR)$. This formula requires a physical noise model (the SNR parameter) and is a theorem about the maximum information rate through a continuous Gaussian channel. Neither the physical noise model nor the communication channel exists in nothing's structural space. The formula cannot be derived from nothing's structure, and it is not used.

The Triadic Noise Floor $\sigma > 1/3$ (Constraint II, Essay I) is present in Essay III, but derived correctly: as the condition that the Boundary-axis clearance fraction exceeds the degenerate uniform value $1/3$. This is a pure lattice-arithmetic condition, not a signal-to-noise ratio. Similarly, the Minimum Bit-Budget $I\_min = 0.2$ is derived from the two-step minimum for binary distinction on the discrete lattice — not from a channel-capacity formula.

What Essay III Adds

Seven genuinely new results, all derived from counting on the Essay II lattice:

T.FA (Face-Addressing Theorem): Three faces require between *1* and *2* bits to address; the forbidden maximum is *log2(3) ≈ 1.585 bits* — achieved only in the degenerate uniform case forbidden by T.NE.

T.FE (Face-Entropy Theorem): The actual face-label entropy is strictly below *log2(3)* — the informational expression of T.NE.

T.AH (Axis-Hierarchy Theorem): $H_B > H_R > H_S$ — axis-entropy ordering matches clearance ordering exactly.

T.CMA (Clearance Margin Addressing): *Source = 1 bit*, *Boundary = 2 bits*, *Remainder = 1.577 bits* (non-integer). The Boundary margin is exactly one bit richer than the Source margin.

T.TSI (Triadic Sub-Volume Information): *log2(1/TI) = log2(125/42) ≈ 1.574 bits*, which is *log2(3) minus log2(126/125)* — the informational expression of *TI* exceeding *1/3* by *1/375*.

T.DCI (Dimensional Collapse Information): The dimensional collapse preserves exactly *1/d* = *1/3* of the full coordinate information — the same ratio as the continuum exponent.

T.SPI (Structural Pixel Information): *log2(25) ≈ 4.64 bits* to address a specific Structural Pixel. The Ontological Shadow is the minimum-probability 2D clearance cell.

Part I — Addressing Nothing's Structural Space

The first informational question is the simplest: given the discrete lattice of nothing's structural space, how many bits are required to specify the address of any position? This is a question of pure counting: count the positions, take the base-2 logarithm, round up to the nearest integer. The answer is an exact property of the lattice derived in Essays I and II.

I.1 The Bit-Definition

Before computing any information quantities, the bit must be defined in terms that are internal to nothing's structural space.

> **Definition** D.BIT · **The Bit as Binary Address on the Discrete Lattice**
>
> A bit is the minimum unit of binary address — the information required to make a single binary distinction: between position A and position B, when exactly two positions are addressable.
>
> On the $n = 10$ discrete lattice of Essay II, a bit arises naturally: the lattice grain $\delta = 1/10$ is the minimum addressable unit. Any two adjacent lattice positions — differing by exactly δ — require exactly one binary choice to distinguish: 'is the coordinate at position i or position i+1?' Each such binary choice is one bit.
>
> ```
> 1 bit  =  the information required to distinguish
> between 2 adjacent lattice positions
> ```
> (1)
>
> This definition is internal to the lattice. It requires no external communication model. The bit is the combinatorial unit of the discrete lattice — nothing more and nothing less.

The bit defined above is the same object as the binary logarithm: if there are N distinct positions, the number of bits required to address any one of them is $log2(N)$. This is because addressing N positions requires a binary string long enough to enumerate all N — which requires *ceiling(log2(N))* binary digits. The exact (non-integer) value *log2(N)* is the minimum bit-count; the integer ceiling is the minimum whole-bit count.

Principle P.BA · **The Binary Addressing Principle**

For any finite set of N distinct positions:

```
Exact addressing bits:  log2(N)
(the exact minimum bits)                                    (2)

Integer addressing bits: ceil(log2(N))
(the minimum whole bits)                                    (3)
```

This principle is used throughout Essay III. Every information quantity is derived by counting positions and applying *log2*. No probability distribution is imposed externally — when fractions arise, they are ratios of position-counts to total-counts, derivable from the lattice geometry of Essay II.

I.2 Per-Axis Addressing

Each axis of nothing's structural space *NS* has *n = 10* addressable positions (V.DS, Essay I; D.LS, Essay II). Addressing any specific position on one axis requires:

Derivation V.PAA · **Per-Axis Addressing Bits**

One axis: *n = 10* positions.

```
Exact bits:    log2(10)  =  log2(2) + log2(5)  =
  1 + log2(5)  =  3.3219 bits                               (4)

Integer bits:  ceil(log2(10))  =  ceil(3.3219)  =
  4 bits                                                    (5)
```

Verification: $2^3 = 8 < 10 \leq 16 = 2^4$. Therefore 3 bits are insufficient (can only address 8 positions); 4 bits are necessary and sufficient (can address up to 16 positions, using 10 of the 16 available codes).

The fractional excess: $ceil(log2(10)) - log2(10) = 4 - 3.3219 = 0.6781$ *bits*. This means 4 bits can address 16 positions, but only 10 are needed. The remaining 6 codes ($16 - 10 = 6$) are structurally unoccupied address codes on each axis — a consequence of $n = 10$ not being a power of 2.

I.3 Full Lattice Addressing

The full three-dimensional lattice NS has three independent axes, each requiring its own address. Because the axes are independent (T.GA, Essay II), their addresses are independent — each axis-address requires no information about the other axes.

Derivation V.FLA · **Full Lattice Addressing Bits**

Full NS: 10 x 10 x 10 = 1000 positions.

```
Exact bits:    log2(10^3)  =  3 x log2(10)  =
3 x 3.3219  =  9.9658 bits
```
(6)

```
Integer bits:  ceil(9.9658)  =  10 bits
```
(7)

Verification: $2^9 = 512 < 1000 \leq 1024 = 2^{10}$. Ten bits are necessary and sufficient.

Independent-axis decomposition: because the three axes are independent, the 10-bit address decomposes into three 4-bit axis-addresses minus 2 bits of overlap (since $4+4+4=12 > 10$):

```
Full address = 10 bits  <  3 x 4 bits = 12 bits
(axes are not entirely independent in integer-bit
representation)
```
(8)

> The exact decomposition is *3 x log2(10) = 9.9658 bits* — exactly 3 times the per-axis information. The integer ceiling introduces a small efficiency loss. The exact value *9.9658* is the precise informational content of nothing's complete structural address.

I.4 The Three-Face Address Requirement

Essay I (T.TR) established that the Triad has exactly three faces. Essay II (D.3A) assigned each face its own axis. What is the minimum number of bits required to address the face-label — to specify which of the three faces a logical position belongs to?

> **Theorem** T.FA · **The Face-Addressing Theorem**
>
> The three faces of the Triad — Source, Boundary, Remainder — are three distinct labels. To specify one of three labels requires:
>
> ```
> Exact face-address bits:   log2(3)  =  1.5850 bits          (9)
>
> Integer face-address bits: ceil(log2(3))  =  2 bits         (10)
> ```
>
> This is the minimum bit-count for any binary addressing system that can distinguish between the three faces. It requires at most 2 binary digits (since *2^2 = 4 >= 3*). Exactly 1 bit is insufficient (*2^1 = 2 < 3* — cannot distinguish three faces with one bit).
>
> The exact value *log2(3) = 1.5850 bits* is the face-label information content assuming equal face sizes (uniform distribution over the three faces). In the actual Triad, face sizes are unequal (T.NE, Essay I). Therefore *log2(3)* is not the actual face-label entropy but the maximum it could achieve — and this maximum is forbidden.

Part II — The Face-Entropy and the Forbidden Ceiling

Part II derives the informational expression of T.NE (Essay I): the three faces are strictly unequal in size. In counting terms: the three face-territories of *NS* have unequal numbers of positions. This unequal distribution reduces the face-label entropy below its maximum $log2(3)$. The maximum — $log2(3) \approx 1.585$ *bits* — is the information content of the degenerate uniform three-face partition, which T.NE explicitly forbids. It is the Forbidden Ceiling.

II.1 The Face-Territory Sizes

The three faces of the Triad each own a specific territory in *NS*. But it is important to be precise about what 'face-territory' means in the three-dimensional lattice. Each face has its own AXIS; the face-territory in *NS* is the slab of *NS* where the face's own-axis coordinate is in the face's clearance margin.

For a position in *NS* to 'belong to the Source face,' its Source-axis coordinate x_S must be in the Source clearance margin $SM = [E, 1] = [4/5, 1]$. Similarly for Boundary (x_B in *BM*) and Remainder (x_R in *RM*). But these are three separate binary classifications — a position can be in none, one, two, or all three clearance territories simultaneously. The face-label is not a single three-way classification of *NS*; it is three independent binary classifications, one per axis.

The correct informational treatment therefore operates axis by axis: on each axis, there is a binary partition into primitive territory and clearance territory. The entropy of this binary partition is the axis-entropy — a value between 0 and 1 bit, representing the informational content of the clearance-label on that axis.

Definition D.AE · **The Axis-Entropy**

The axis-entropy of a face-axis is the binary entropy of the clearance-label on that axis: $H(p) = -p\ log2(p) - (1-p)\ log2(1-p)$, where p is the fraction of axis positions in the clearance margin.

$$H\_S = H(I\_min) = H(1/5) = -(1/5)\ log2(1/5) - (4/5)\ log2(4/5) \tag{11}$$

$$H\_B = H(\sigma) = H(2/5) = -(2/5)\ log2(2/5) - (3/5)\ log2(3/5) \tag{12}$$

$$H\_R = H(\Lambda) = H(0.2984) = -0.2984\ log2(0.2984) - 0.7016\ log2(0.7016) \tag{13}$$

Each axis-entropy H measures how much information is contained in knowing whether a position on that axis is in the clearance or not. $H = 0$ means the partition is completely predictable (all positions on one side); $H = 1$ bit means the partition is maximally uncertain (exactly half the positions on each side).

II.2 Computing the Three Axis-Entropies

Derivation V.AEC · **The Three Axis-Entropies: Exact Computation**

Source axis: clearance fraction $I\_min = 1/5 = 0.2$ (2 positions in clearance, *8* in primitive).

$$H\_S = -(1/5)\ log2(1/5) - (4/5)\ log2(4/5) \tag{14}$$

$$= (1/5)\ x\ 2.3219 + (4/5)\ x\ 0.3219 \tag{15}$$

$$= 0.4644 + 0.2575 \tag{16}$$

$$= 0.7219\ bits \tag{17}$$

Boundary axis: clearance fraction $\sigma = 2/5 = 0.4$ (4 positions in clearance, *6* in primitive).

$$H\_B = -(2/5)\ log2(2/5) - (3/5)\ log2(3/5) \tag{18}$$

$$= (2/5)\ x\ 1.3219 + (3/5)\ x\ 0.7370 \tag{19}$$

$$= 0.5288 + 0.4422 \tag{20}$$

$$= 0.9710\ bits \tag{21}$$

Remainder axis: clearance fraction $\Lambda = 0.2984$ (*2.984* positions in clearance, *7.016* in primitive).

$$H\_R = -0.2984 \log_2(0.2984) - 0.7016 \log_2(0.7016) \tag{22}$$

$$= 0.2984 \times 1.7443 + 0.7016 \times 0.5122 \tag{23}$$

$$= 0.5204 + 0.3594 \tag{24}$$

$$= 0.8793 \text{ bits} \tag{25}$$

II.3 The Degenerate Ceiling

What would the axis-entropy be in the degenerate case that T.NE forbids — where each clearance is exactly *1/3* (equal partition of each axis)?

Derivation V.DEC · **The Degenerate Axis-Entropy**

Degenerate case: clearance fraction = 1/3 on every axis (forbidden by T.NE via T.IC):

$$H\_degenerate = H(1/3) = -(1/3) \log_2(1/3) - (2/3) \log_2(2/3) \tag{26}$$

$$= (1/3) \times 1.5850 + (2/3) \times 0.5850 \tag{27}$$

$$= 0.5283 + 0.3900 \tag{28}$$

$$= 0.9183 \text{ bits} \tag{29}$$

The degenerate axis-entropy *H_degenerate = 0.9183* bits is the axis-entropy value that would arise if every clearance were exactly *1/3* of its axis. This value is forbidden because T.IC (Essay I) established that the lattice grain $n = 10$ must be incommensurate with *3* (*gcd(n,3) = 1*), ensuring that *1/3* is not a lattice address. The degenerate state cannot be stably occupied.

II.4 T.FE — The Face-Entropy Theorem

Theorem T.FE · **The Face-Entropy Theorem**

The three axis-entropies satisfy:

$$H\_B = 0.9710 \text{ bits} > H\_degenerate = 0.9183 \text{ bits} > H\_S = 0.7219 \text{ bits} \tag{30}$$

$$H\_R = 0.8793 \text{ bits} \quad (\text{intermediate: } H\_B > H\_R > H\_S) \tag{31}$$

Implications:

(i) ***H_B > H_degenerate***: the Boundary axis is informationally RICHER than the degenerate case. The Boundary clearance $\sigma = 2/5 = 0.4$ is closer to *0.5* (maximum-entropy partition) than *1/3* is. The Boundary axis provides more information per position than the degenerate partition would.

(ii) ***H_S < H_degenerate***: the Source axis is informationally POORER than the degenerate case. The Source clearance $I\_min = 1/5 = 0.2$ is farther from *0.5* than *1/3* is. The Source axis provides less information per position — consistent with P.HI (Essay I): the Source face is the concentrated reservoir, not the highly differentiated frontier.

(iii) ***H_R is intermediate***: $\Lambda = 0.2984 \approx 0.3$ lies between $1/3 = 0.333$ and $1/5 = 0.2$, giving *H_R between H_degenerate and H_S*.

The axis-entropy ordering $H\_B > H\_R > H\_S$ matches exactly the clearance ordering $\sigma > \Lambda > I\_min$, which matches exactly the structural hierarchy P.HI. The informational register confirms and expresses the geometric and ontological hierarchy from one unified counting perspective.

II.5 The Forbidden Ceiling: *log2(3)* as the Informational Expression of T.NE

T.FA established that the maximum face-label entropy is $log2(3) \approx 1.585$ *bits* — the information content of a uniform distribution over three faces. T.NE (Essay I) established that the three faces are strictly unequal. Together these give the Forbidden Ceiling theorem.

Theorem T.FC · **The Forbidden Ceiling**

The maximum axis-entropy is $H\_max = H(0.5) = 1$ *bit* (exactly half the axis in clearance, half in primitive). The face-entropy of a uniform THREE-face labelling is $log2(3) = 1.5850$ *bits*. Neither is achievable:

$H\_max = 1$ bit: achievable only if each axis clearance = $1/2$ of the axis (5 out of 10 positions). But none of the three clearances is $1/2$: $I\_min = 0.2$, $\sigma = 0.4$, $\Lambda = 0.2984$. The maximum-entropy axis partition is not realised.

$log2(3) = 1.585$ bits: the entropy of choosing one of three equally-sized faces. Achievable only in the degenerate uniform case where each face occupies exactly $1/3$ of the lattice. This is T.NE's forbidden state — the equal partition where Source = Boundary = Remainder in measure. T.NE (Essay I) proved that equality of any two faces destroys the Triad.

$$\text{The Forbidden Ceiling: } H\_face\_actual < log2(3) = 1.5850 \text{ bits} \tag{32}$$

log2(3) is the informational upper bound for the three-face entropy that cannot be reached without destroying the Triad. It is not an achieved information content — it is the ceiling that nothing's structural inequality (T.NE) keeps nothing permanently below.

Part III — The Minimum Informational Content of Each Clearance

Part III asks: how many bits are required to address a specific position within each clearance margin? This is the address-space of each clearance: how many distinct positions it contains, and what their addressing costs are. The results are exact, integer for Source and Boundary, and non-integer for Remainder.

III.1 The Minimum Binary Distinction

The most fundamental informational result follows directly from the definition of a bit (D.BIT). To make any binary distinction — to distinguish 'inside clearance' from 'outside clearance' — requires exactly *1 bit*. But to distinguish specific positions WITHIN the clearance from each other requires additional bits: *log2(k) bits* for *k* positions.

The minimum clearance that can contain any informational content is a clearance with at least *2* positions. A single-position clearance has *log2(1) = 0 bits* — it carries no information because there is nothing to distinguish within it. The minimum for any informational content is therefore *2* positions: *1* bit to distinguish which of the *2* positions within the clearance a coordinate falls on.

Theorem T.MBD · **The Minimum Binary Distinction**

A clearance margin contains informational content if and only if it has at least 2 addressable positions

.

```
1 position:
log2(1) = 0 bits  (no distinction possible — zero
information)                                          (33)

2 positions:
log2(2) = 1 bit   (minimum binary distinction)        (34)
```

```
4 positions:
log2(4) = 2 bits  (two binary distinctions)                    (35)

k positions:
log2(k) bits  (exact addressing)                               (36)
```

The minimum viable clearance is 2 positions. This is the information-theoretic necessity of *I_min = 2 x δ*: a Source clearance of only 1 position *(I_min = δ = 0.1)* would have *log2(1) = 0 bits* — no internal structure, nothing to distinguish within it, a clearance that carries no information. The minimum for a clearance with informational content is 2 positions.

III.2 The Source Clearance: Exactly *1 Bit*

The Source clearance margin *SM = [E, 1] = [4/5, 1]* contains exactly 2 addressable positions on the *n = 10* lattice: *x_S = 8/10* and *x_S = 9/10* (the two lattice steps in the Source clearance, corresponding to a_steps = 2, established in V.MB, Essay I).

Theorem T.SC1 · **Source Clearance: Exactly *1 Bit***

Source clearance contains 2 lattice positions.

```
I_min = 2 x δ  =>
2 lattice positions in Source clearance                        (37)

Addressing bits:
log2(2)  =  1 bit  (exact)                                     (38)
```

This *1 bit* is the informational content of the Source clearance: a single binary distinction between the two Source-clearance positions. The positions are *{x_S = 8/10, x_S = 9/10}*.

Informational necessity of 2 positions: from T.MBD, a 1-position Source clearance would carry *0 bits* — no internal structure. The Source face requires internal structure: it is the reservoir of indetermination (D.3F, Essay I), not a boundary marker. Internal structure requires at least 2 distinct positions. Therefore the Source clearance must have at least 2 positions — consistent with *I_min = 2 x δ* forced by V.MB.

The *1-bit* Source clearance is the MINIMUM possible for a structurally functional clearance. It is the tightest the Source margin can be while still carrying any informational content.

III.3 The Boundary Clearance: Exactly 2 *Bits*

The Boundary clearance margin $BM = [F, 1] = [3/5, 1]$ contains exactly 4 addressable positions on the $n = 10$ lattice: $x_B = 6/10, 7/10, 8/10, 9/10$ (the four lattice steps, $c_{steps} = 4$, forced by the noise-floor condition $c > n/3$).

Theorem T.BC2 · **Boundary Clearance: Exactly 2 Bits**

Boundary clearance contains 4 lattice positions.

```
σ = 4 x δ  =>
4 lattice positions in Boundary clearance                    (39)

Addressing bits:
log2(4)  =  log2(2^2)  =  2 bits  (exact)                    (40)
```

This *2-bit* Boundary clearance has an important structural interpretation. Two bits can distinguish 4 positions. These 4 positions can be organised as a 2-level binary tree: at the first level, 2 positions; at the second level, 2 more positions. The Boundary clearance is the first clearance where a complete second binary distinction is possible within the margin.

The *2-bit* Boundary clearance is not accidental — it is forced by $c = 4 = 2^2$ positions (T.UC, Essay I). The step count $c = 4$ was derived as the minimum integer above $n/3 = 10/3 = 3.33$, meaning $c = 4$ is the minimum integer satisfying $c > n/3$. And $4 = 2^2$ is the smallest power of 2 above 3, giving exactly *2 bits*.

III.4 The One-Bit Step: The Informational Expression of $c/a = 2$

The Source clearance has *1 bit*; the Boundary clearance has *2 bits*. The difference is exactly *1 bit*. This one-bit step is the informational expression of the Unique Combinatorial Lock's ratio $c/a = 4/2 = 2$ (T.UC, Essay I).

Theorem T.BOS · **The One-Bit Step Between Source and Boundary**

From T.SC1 and T.BC2:

$$\text{Boundary addressing bits} - \text{Source addressing bits} = 2 - 1 = 1 \text{ bit} \tag{41}$$

Derivation: the step-count ratio $c/a = 4/2 = 2 = 2^1$. Taking $\log_2$ of both sides:

$$\log_2(c/a) = \log_2(4/2) = \log_2(4) - \log_2(2) = 2 - 1 = 1 \text{ bit} \tag{42}$$

This 1-bit step is forced by $c/a = 2$, which is forced by T.UC (Essay I): the minimum even-sum constraint requires $a + c$ even, the minimum step-count constraint requires $c >= 4$, and the minimality principle of T.PS gives the smallest $a = 2$. These together force $c = 2a$, giving a bit-step of $\log_2(c/a) = \log_2(2) = 1$ *bit*.

The Boundary clearance always carries exactly one bit more than the Source clearance. This is the informational hierarchy at the clearance level — not just that the Boundary is larger, but that it is precisely one binary step richer.

III.5 The Remainder Clearance: Non-Integer Bits

The Remainder clearance margin $RM = [\delta, 1] = [0.7016, 1]$ contains $\Lambda/\delta = 0.2984/0.1 = 2.984$ lattice positions — a non-integer count. This is the first clearance whose addressing bits are not a whole number.

Theorem T.RC · **Remainder Clearance: Non-Integer Bits**

Remainder clearance contains 2.984 effective lattice positions.

```
Λ / δ  =  0.2984 / 0.1  =  2.984 positions
```
(43)

```
Exact addressing bits:  log2(2.984)  =  1.5772 bits
```
(44)

```
Integer ceiling:  ceil(1.5772)  =  2 bits
```
(45)

The non-integer character of the Remainder clearance bits is the informational expression of the sub-lattice residual derived in Essay II (V.RMG): $\Lambda = 0.2984 \neq k \times \delta$ for any integer k. The lower boundary of RM falls at $x\_R = \delta = 0.7016$, which is between lattice positions $7/10$ and $8/10$.

Addressing consequence: *2 bits* can address 4 positions. The Remainder clearance uses only *2.984* of those 4 address codes — leaving $4 - 2.984 = 1.016$ address codes structurally unused. This 'wasted' addressing capacity corresponds to the sub-lattice residual $\varepsilon\_R = 0.0016$ (the amount by which δ falls above the nearest whole-step boundary $7/10 = 0.7000$).

III.6 The Sub-Lattice Information

The non-integer Remainder clearance introduces a concept absent from the Source and Boundary clearances: sub-lattice information. This is the informational content of the partial first step of the Remainder clearance — the portion of the step from x_R = 7/10 to x_R = δ = 0.7016 that the lattice cannot address as a full step.

Definition D.SLI · **Sub-Lattice Information**

The sub-lattice residual $\varepsilon\_R = \delta - floor(\delta/\delta) \times \delta = 0.7016 - 0.7000 = 0.0016$ (V.RMG, Essay II) represents the fractional first step of the Remainder clearance. As a fraction of one lattice step:

```
ε_R / δ  =  0.0016 / 0.1  =  0.016
(1.6% of one lattice step)
```
(46)

The information content of this fractional step:

$$I(\text{sub-lattice}) = -\log_2(0.016) = 5.966 \text{ bits} \tag{47}$$

This *5.966 bits* is the information required to locate a position within the sub-lattice fraction — if such a position were addressable. But on the $n = 10$ discrete lattice, sub-δ positions are structurally unaddressable. The sub-lattice information is therefore the information content of a distinction the lattice CANNOT make: the information that is permanently expelled by the lattice's own resolution. It is the informational analog of the snap residual $\varepsilon_{snap} = 1/30$ from the Structural Verification of Essay I.

III.7 The Three Clearances Summarised as Bit-Addresses

Principle P.CBA · **The Three Clearance Bit-Addresses**

Summary of addressing bits for the three clearance margins:

```
Source clearance:    log2(2)= 1.0000 bits
(exact integer — 2 lattice positions)                    (48)

Boundary clearance:  log2(4)= 2.0000 bits
(exact integer — 4 lattice positions)                    (49)

Remainder clearance: log2(2.984) = 1.5772 bits
(non-integer — 2.984 effective positions)                (50)
```

Structural interpretation:

Source: *1 bit* = minimum informational content for any structurally functional clearance. The Source is at the information floor.

Boundary: *2 bits* = one binary step above the Source. The Boundary is richer by exactly *1 bit*, forced by $c/a = 2$ (T.BOS).

Remainder: *1.577 bits* = between Source and Boundary. Not integer because the Remainder margin arises from a three-dimensional computation (dimensional collapse) rather than a direct lattice-step count.

```
Ordering:

Boundary (2.000) > Remainder (1.577) > Source (1.000)
```
(51)

This ordering matches the clearance ordering $\sigma > \Lambda > I\_min$, and the axis-entropy ordering $H\_B > H\_R > H\_S$, and the structural hierarchy P.HI. The informational register is consistent with the geometric and ontological registers.

Part IV — The Axis-Entropy Hierarchy and Its Structural Meaning

Part III established the bit-addresses of the three clearance margins as position-counts. Part IV establishes the axis-entropies as binary-partition measures — measuring not just how many positions are in the clearance, but the full information-theoretic character of the clearance/primitive partition on each axis. These two measures (bit-address and axis-entropy) agree in their ordering but differ in their absolute values and their geometric meaning.

IV.1 The Axis-Entropy as Partition Measure

The axis-entropy $H\_X = H(p, 1\text{-}p)$ where p = *(clearance positions)/(total positions) = clearance fraction* is a different measure from the bit-address *log2(clearance positions)*. The bit-address measures the information needed to locate a position WITHIN the clearance. The axis-entropy measures the information in the clearance/primitive classification ITSELF — how much is learned by knowing whether a coordinate is in the clearance or not.

Definition D.AEM · **The Two Informational Measures Contrasted**

For each axis:

Bit-address (*log2(k)*): how many bits to locate a specific position WITHIN the clearance, given k clearance positions.

Axis-entropy H(*p, 1-p*): how many bits are learned by being told whether a RANDOM position on the axis is in the clearance or not, given clearance fraction p.

These are different. The bit-address depends only on the clearance SIZE (number of positions). The axis-entropy depends on the RATIO between clearance and non-clearance — it measures the binary partition's balance.

$$\text{Bit-address} = \log_2(k) \quad \text{where } k = \text{clearance positions} \tag{52}$$

$$\text{Axis-entropy} = -p \log_2(p) - (1-p) \log_2(1-p) \quad \text{where } p = k/n \tag{53}$$

Both are maximum at the 50/50 partition: bit-address is $log2(n/2) = log2(5) = 2.32$ *bits*; axis-entropy is $H(0.5) = 1.00$ *bit*. The axis-entropy's maximum of 1 bit is always lower than the bit-address's maximum $log2(n) = log2(10) = 3.32$ *bits*.

IV.2 The Axis-Entropy Values and Their Derivation

Derivation V.AEH · **The Axis-Entropy Hierarchy: Full Derivation**

Source axis: $p\_S = I\_min = 1/5$.

$$H\_S = -(1/5)\ log2(1/5) - (4/5)\ log2(4/5) = 0.7219\ bits \tag{54}$$

Boundary axis: $p\_B = \sigma = 2/5$.

$$H\_B = -(2/5)\ log2(2/5) - (3/5)\ log2(3/5) = 0.9710\ bits \tag{55}$$

Remainder axis: $p\_R = \Lambda = 0.2984$.

$$H\_R = -0.2984\ log2(0.2984) - 0.7016\ log2(0.7016) = 0.8793\ bits \tag{56}$$

Degenerate case: $p\_deg = 1/3$.

$$H\_deg = -(1/3)\ log2(1/3) - (2/3)\ log2(2/3) = 0.9183\ bits \tag{57}$$

Ordering:

$$H\_B = 0.9710 > H\_deg = 0.9183 > H\_R = 0.8793 > H\_S = 0.7219 > 0 \tag{58}$$

> The Boundary axis is the only axis with entropy exceeding the degenerate value. This is the informational content of T.DC.III (Essay I): the Boundary clearance is precisely calibrated to be informationally richer than the degenerate partition — it is the most 'uncertain' axis classification in nothing's structural space.

IV.3 Why H_B Exceeds the Degenerate Value

The degenerate value $H\_deg = H(1/3) = 0.9183$ *bits* arises from $p\_deg = 1/3$. The binary entropy function $H(p)$ is maximised at $p = 0.5$ (giving $H = 1$ *bit*) and decreases symmetrically toward 0 as $p \to 0$ or $p \to 1$. For $p < 0.5$, $H(p)$ is increasing in p: more clearance (up to 50%) gives more information.

$H\_B > H\_deg$ because $\sigma = 0.4$ is closer to 0.5 than $1/3$ is. Numerically: $|\sigma - 0.5| = |0.4 - 0.5| = 0.1$, while $|1/3 - 0.5| = 1/6 \approx 0.167$. The Boundary clearance is closer to the maximum-entropy partition than the degenerate value is.

Theorem T.BD · **The Boundary Axis Discriminability Theorem**

The Boundary clearance $\sigma = 2/5 = 0.4$ satisfies:

$$\sigma = 2/5 \quad \text{in} \quad (1/3,\ 1/2) \tag{59}$$

This range means σ is strictly between the degenerate value $(1/3)$ and the maximum-entropy partition $(1/2)$. Therefore:

$$H\_deg = H(1/3) < H(\sigma) = H\_B < H(1/2) = 1 \text{ bit} \tag{60}$$

$$0.9183 < 0.9710 < 1.0000 \tag{61}$$

The Boundary axis is informationally discriminable from the degenerate state: $H\_B \neq H\_deg$. This is the informational expression of the constraint C.II (Essay I): $\sigma > 1/3$. The Boundary clearance was forced above $1/3$ to prevent the degenerate state from being occupiable. In informational terms: the Boundary axis must be MORE informative than the degenerate partition — and it is.

IV.4 The Joint Clearance-Label Entropy

The three axis-entropies measure the information content of each axis's clearance-label independently. Because the three axes are independent (T.GA, Essay II), the joint information of all three clearance-labels is the sum of the three individual entropies.

Theorem T.JCE · **The Joint Clearance-Label Entropy**

Since the three axes are independent:

$$H\_joint = H\_S + H\_B + H\_R = 0.7219 + 0.9710 + 0.8793 = 2.5722 \text{ bits} \tag{62}$$

Compare to limiting cases:

$$\text{Maximum (all axes at } p = 0.5\text{):} \quad H\_max\_joint = 3 \times 1.0000 = 3.0000 \text{ bits} \tag{63}$$

$$\text{Degenerate (all axes at } p = 1/3\text{):} \quad H\_deg\_joint = 3 \times 0.9183 = 2.7549 \text{ bits} \tag{64}$$

Actual:

$$H\_joint = 2.5722 \text{ bits} \tag{65}$$

The actual joint entropy (*2.5722 bits*) is below the degenerate value (*2.7549 bits*). Nothing's clearance architecture is informationally MORE concentrated than the degenerate state — each axis is more 'decided' (further from 50/50) than the degenerate partition. This is a global expression of T.NE: the unequal face-sizes reduce the total joint entropy below the degenerate ceiling.

Efficiency: *H_joint / H_max_joint = 2.5722 / 3.0000 = 0.8574 = 85.7%*. Nothing's clearance architecture uses *85.7%* of the maximum possible joint entropy of its three axes. It is informationally rich but not maximally uncertain.

IV.5 The Failure Modes as Information Boundaries

The three structural failure modes derived in Essays I and II each correspond to an informational boundary condition — a limit at which one axis-entropy approaches a degenerate value.

Principle P.FMI · **The Failure Modes as Informational Limits**

Deterministic Crystal *(σ -> 0, F -> 1)*:

```
H_B -> H(0) = 0 bits
(Boundary axis loses all clearance information)
```
(66)

The Boundary clearance vanishes: nothing to distinguish within it

Total Dissolution *(σ -> 1, F -> 0)*:

```
H_B -> H(1) = 0 bits
(Boundary axis again loses all clearance information)
```
(67)

The Boundary clearance fills the entire axis: primitive territory vanishes

Causal Closure Seizure *(Λ -> 0, δ -> 1)*:

```
H_R -> H(0) = 0 bits
(Remainder axis loses all clearance information)
```
(68)

The Remainder margin vanishes: no informational residual remains

Structural observation: both extremes of *σ* (*0* and *1*) drive *H_B* to *0*. The axis-entropy is zero at both failure modes. The maximum *H_B = 0.9710 bits* at *σ = 0.4* is the unique structural maximum — the point furthest from both failure modes simultaneously.

Part V — The Triadic Sub-Volume and Its Informational Character

The Triadic Sub-Volume (*TSV = [0,E] x [0,F] x [0,C]*) was defined geometrically in Essay II (D.TSV) as the three-dimensional occupied region of *NS*, with volume *TI = E x C x F = 42/125 = 0.336*. Part V reads *TSV* in the informational register: how much information is contained in knowing that a lattice position is inside *TSV*, and how does this information relate to the Forbidden Ceiling *log2(3)*?

V.1 The TSV Position Count

On the *10 x 10 x 10* discrete lattice, the interior positions (strictly between *0* and *1* on all axes) number *9^3 = 729*. Among these, the *TSV* occupies the following interior positions: *x_S* in *{1/10, ..., 7/10}* (7 of 9 interior Source-axis positions at or below *E = 8/10*), *x_B* in *{1/10, ..., 5/10}* (5 of 9 interior Boundary-axis positions at or below *F = 6/10*), *x_R* in *{1/10, ..., 6/10}* (6 of 9 interior Remainder-axis positions at or below *C = 7/10*).

However, for the purpose of counting positions in *TSV* vs. *NS*, it is cleaner to count axis positions from *0* through *9* (the 10 lattice steps on each axis, from position *0* to position *9/10*):

Derivation V.TSVP · **TSV Position Count**

On the *10*-position-per-axis lattice (positions *0* through *9/10* = steps *0* through *9*):

```
Source axis:    positions 0 to E x n - 1  =  0 to 7   =>
   8 positions                                          (1)

Boundary axis:  positions 0 to F x n - 1  =  0 to 5   =>
   6 positions                                          (2)

Remainder axis: positions 0 to C x n - 1  =  0 to 6   =>
   7 positions                                          (3)

TSV positions  =  8  x  6  x  7  =  336                 (4)
```

$$\text{Total } NS \text{ positions} = 10^3 = 1000 \tag{5}$$

$$TSV \text{ fraction} = 336 / 1000 = 0.336 = TI \text{ ok} \tag{6}$$

V.2 T.TSI — The Triadic Sub-Volume Information

Theorem T.TSI · **The Triadic Sub-Volume Information**

The information content of knowing that a random *NS* position is inside *TSV*:

$$I(TSV) = -\log2(336/1000) = -\log2(0.336) = \log2(1000/336) = \log2(125/42) \tag{7}$$

$$\log2(125/42) = \log2(125) - \log2(42) \tag{8}$$

$$= \log2(5^3) - \log2(2 \times 3 \times 7) \tag{9}$$

$$= 3 \log2(5) - (\log2(2) + \log2(3) + \log2(7)) \tag{10}$$

$$= 3 \times 2.3219 - (1 + 1.5850 + 2.8074) \tag{11}$$

$$= 6.9658 - 5.3924 \tag{12}$$

$$= 1.5735 \text{ bits} \tag{13}$$

Compare to the Forbidden Ceiling:

$$I(TSV) = 1.5735 \text{ bits} < \log2(3) = 1.5850 \text{ bits} \quad \text{(below Forbidden Ceiling)} \tag{14}$$

The information content of the *TSV* is *0.0115* bits below the Forbidden Ceiling.

V.3 Why *I(TSV)* is Below *log2(3)*

The proximity of *I(TSV) = log2(125/42)* to *log2(3) = log2(126/42)* is not a numerical coincidence. It follows from the exact value of *TI* and its relationship to *1/3*.

Derivation V.TSVC · **The Structural Relationship Between I(TSV) and log2(3)**

Compare *TI = 42/125* to *1/3*:

$$1/3 = 125/375 = 41.667.../125 \tag{15}$$

$$TI = 42/125 \tag{16}$$

$$TI - 1/3 = 42/125 - 1/3 = 126/375 - 125/375 = 1/375 \tag{17}$$

TI exceeds *1/3* by exactly *1/375*. This means *TI = 1/3 + 1/375*, so:

$$I(TSV) = \log2(1/TI) = \log2(1/(1/3 + 1/375)) \tag{18}$$

$$= \log2(375/126) = \log2(125/42) \tag{19}$$

The ratio to *log2(3)*:

$$\log2(3) - I(TSV) = \log2(3) - \log2(125/42) = \log2(3 \times 42/125) = \log2(126/125) \tag{20}$$

$$= \log2(1.008) = 0.0115 \text{ bits} \tag{21}$$

The deficit below the Forbidden Ceiling is *log2(126/125)*. This is fully determined by the excess of *TI* over *1/3*, which is *1/375*. Both are consequences of T.NE: the three face sizes are unequal, making the *TSV* slightly larger than the degenerate *1/3* value.

V.4 The Structural Meaning of TI Exceeding ⅓

$TI = 42/125 > 1/3$ means the Triadic Sub-Volume occupies MORE than one-third of nothing's structural space. This is a direct consequence of the actual primitives being forced (by T.UC) to have an average larger than the degenerate $2/3$:

Derivation V.TSVM · **Why $TI > (1/3)^3$ and $TI > 1/3$**

Average of the three primitives:

$$(E + C + F) / 3 = (4/5 + 7/10 + 3/5) / 3 = (8/10 + 7/10 + 6/10) / 3 = 21/30 = 7/10 \tag{22}$$

Degenerate case: each primitive = $1 - 1/3 = 2/3 \approx 0.667$. Actual average: $7/10 = 0.700 > 0.667$.

The actual primitives have a HIGHER average than the degenerate case. The Triad's non-equal face sizes (T.NE), combined with the minimality principle (T.PS: minimum clearances), produce primitives that are on average larger than the degenerate $2/3$.

$$\text{Degenerate } TI: (2/3)^3 = 8/27 = 0.296 \tag{23}$$

$$\text{Actual } TI: 42/125 = 0.336 \tag{24}$$

$$\text{Difference: } TI - (2/3)^3 = 0.336 - 0.296 = 0.040 = P = \text{Structural Pixel} \tag{25}$$

A remarkable structural result: the excess of the actual TI over the degenerate TI is exactly equal to the Structural Pixel $P = 1/25 = 0.04$.

V.5 The Clearance Complex Information

The Clearance Complex CC (the complement of TSV in NS) contains $1000 - 336 = 664$ *positions*. Its information content:

Derivation V.CCI · **The Clearance Complex Information**

CC positions: *1000 - 336 = 664*.

$$I(CC) = -\log2(664/1000) = -\log2(0.664) = \log2(1000/664) = \log2(125/83) \tag{26}$$

$$= 0.5867 \text{ bits} \tag{27}$$

Compare to *I(TSV) = 1.5735 bits*:

$$I(TSV) / I(CC) = 1.5735 / 0.5867 = 2.682 \tag{28}$$

The *TSV* is informationally about 2.7 times more surprising than the Clearance Complex — it is less probable (*336* vs. *664* positions), so knowing you are in it tells you more. The Clearance Complex contains the majority of *NS* positions (66.4%), making it the 'expected' territory; the *TSV* is the minority (33.6%), making it informationally significant.

Part VI — The Dimensional Collapse as Information Compression

The dimensional collapse derived in Essay II (T.DC) maps the three-dimensional Triadic Sub-Volume to a single position on the one-dimensional Remainder axis: $\delta = TI^\beta$. Part VI reads this operation in the informational register. The dimensional collapse is an information compression: it reduces a three-axis coordinate to a single-axis coordinate. The amount of information compressed, the fraction preserved, and the relationship of this fraction to the continuum exponent *1/d* are all derivable from the lattice structure.

VI.1 The Full Coordinate Information vs. Single-Axis Information

Derivation V.FCIS · **Full vs. Single-Axis Information**

Full 3D coordinate (any position in *NS*):

```
I(full 3D)  =  log2(10^3)  =  3 x log2(10)  =
9.9658 bits  (integer: 10 bits)
```
(1)

Single Remainder-axis position (any position on *x_R*):

```
I(single axis)  =  log2(10)  =  3.3219 bits
(integer: 4 bits)
```
(2)

Information compressed by dimensional collapse (exact, continuous):

```
I(compressed)  =  I(full 3D) - I(single axis)  =
9.9658 - 3.3219  =  6.6439 bits
```
(3)

Information preserved (fraction):

```
fraction preserved  =  I(single axis) / I(full 3D)  =
3.3219 / 9.9658  =  1/3
```
(4)

> The dimensional collapse preserves exactly *1/3* of the full coordinate information.

VI.2 T.DCI — The Dimensional Collapse Information Theorem

> **Theorem** T.DCI · **The Dimensional Collapse Information Theorem**
>
> The fraction of full coordinate information preserved by the dimensional collapse is exactly *1/d*, where *d = 3* is the Triad's dimension count (T.3D, Essay II).
>
> ```
> Fraction preserved  =  1/d  =  1/3
> ```
> (5)
>
> Proof: The full coordinate information is *d x log2(n)* (*d* independent axes, each with *n* positions). The single-axis Remainder information is *log2(n)*. The ratio is *log2(n) / (d x log2(n)) = 1/d*.
>
> This is the informational expression of the dimensional collapse's geometry. The collapse projects from *d = 3* independent axes to 1 axis, preserving 1 out of *d = 1/3* of the coordinate information. The same ratio *1/d = 1/3* appeared in Essay I as the continuum exponent *β_c = 1/d* — the ideal geometric rate of the dimensional collapse. The continuum exponent and the information-preservation fraction are the same structural object seen from geometric and informational angles simultaneously.

VI.3 The Information Content of the Order Parameter δ

The dimensional collapse maps the three-dimensional *TI* to the one-dimensional δ = TI^β = 0.7016. What is the information content of δ — how much information does the specific value δ = *0.7016* carry within the context of the Remainder axis?

Derivation V.DCI · **The Information Content of δ**

$\delta = 0.7016$ falls between Remainder-axis positions $7/10$ and $8/10$. On the 10-position Remainder axis, positions 0 through 9 correspond to values 0 through $9/10$. $\delta = 0.7016$ is addressed by integer position 7 (the floor).

```
Integer Remainder-axis position of δ:
floor(δ x n) = floor(7.016) = 7                                    (6)

I(position 7 out of 10) = log2(10) = 3.3219 bits                   (7)
```

The information of being at position 7 vs. any other Remainder-axis position: $\log2(10/1) = \log2(10) = 3.32$ *bits* (information gain from specifying one position out of 10). This is the full Remainder-axis address.

The sub-lattice residual: $\delta - 7/10 = 0.7016 - 0.7000 = 0.0016$. As a fraction of one step: $0.0016/0.1 = 0.016$. The information of this sub-lattice specification: $\log2(1/0.016) = 5.97$ *bits* — the information permanently below the lattice's addressable threshold.

VI.4 The Remainder Clearance as a Collapsed Residual

$\Lambda = 1 - \delta = 0.2984$ is the informational residual of the dimensional collapse on the Remainder axis. It is the fraction of the Remainder axis left unoccupied by the collapsed *TSV* projection. Its informational character was established in Part III (T.RC): $\log2(\Lambda/\delta) = \log2(2.984) = 1.577$ *bits* — non-integer, sub-lattice.

The non-integer character of Λ's addressing bits is now fully explained: $\Lambda = 0.2984$ arises from $\delta = TI\hat{}\beta$, and $\beta = 13/40$ is the floor-snap of $1/d = 1/3$. The floor-snap introduces the sub-lattice residual that makes Λ/δ non-integer. If β were exactly $1/d = 1/3$ (the continuum value), δ would be $TI\hat{}(1/3) = (42/125)\hat{}(1/3)$, and Λ would be $1 - TI\hat{}(1/3)$. The floor-snap $\beta = 13/40$ shifts δ by the small amount $\delta\_actual - \delta\_continuum = (42/125)\hat{}(13/40) - (42/125)\hat{}(1/3) \approx 0.7016 - 0.7011 = 0.0005$, which in turn shifts Λ by -0.0005. This sub-δ shift is the informational fingerprint of the floor-snap at Boundary Resolution $R = 40$.

Principle P.RCFI · **The Remainder Clearance as Collapsed Residual**

Λ = 0.2984 is the informational residual of the dimensional collapse. Its non-integer lattice-step count (*2.984 steps*) is the informational expression of the floor-snap *β = 13/40 = floor(R/d)/R*. The floor-snap displaces δ from the continuum value, leaving *Λ* with a non-integer step count. The sub-lattice precision of *Λ* (*0.016* step sub-residual) is informationally inaccessible on the *n = 10* lattice — it is permanently expelled below the lattice's resolution.

VI.5 The Information Preserved vs. Lost in the Collapse

Principle P.ICPL · **Information Preserved and Lost**

The dimensional collapse from 3D to 1D:

```
I(full 3D) = 9.9658 bits (full NS address)                    (8)

I(single axis) = 3.3219 bits (Remainder-axis address)         (9)

I(compressed) = 6.6439 bits
(the two Source/Boundary axes collapsed)                      (10)

Fraction preserved: 1/d = 1/3
(invariant of dimensional collapse)                           (11)
```

The collapsed information (*6.6439 bits*) does not disappear — it is folded into the scalar δ via the power law *δ = TI^β*. The power-law exponent *β = 13/40* governs the rate at which the Source- and Boundary-axis information is folded into the Remainder-axis position. The *1/d* fraction preserved is the structural constant of this folding — it is the same for any three-dimensional system collapsing to one dimension.

Part VII — The Structural Pixel and the Pixel-Address Space

Part VII derives the information content of the Structural Pixel $P = 1/25 = 0.04$ and the Ontological Shadow, using the same counting approach applied throughout. The Structural Pixel is a two-dimensional object in the Source x Boundary projection of *NS*; its information content is the number of bits required to specify which of the *25* non-overlapping pixels a position falls in.

VII.1 The Structural Pixel as an Address Space

The Structural Pixel $P = \sigma \times \delta = 2/5 \times 1/10 = 1/25$ was derived geometrically in Essay II (V.SP, D.SP.II) as the cross-sectional area of the Boundary-margin slab at one Source-axis lattice step. In the Source x Boundary projection of *NS* (ignoring the Remainder axis), the unit square $[0,1] \times [0,1]$ is tiled by exactly *25* non-overlapping Structural Pixels.

Theorem T.SPI · **Structural Pixel Information**

The unit square $[0,1] \times [0,1]$ (Source x Boundary projection of *NS*) contains *25* non-overlapping Structural Pixels, each of area $P = 1/25$.

```
I(pixel address) = log2(25) = log2(5^2) = 2 log2(5) =
4.6439 bits
```
(1)

This is the information required to specify which of the *25* Structural Pixels a position in the Source x Boundary projection falls in. It is the pixel-address space: the 25-way classification of all positions in the 2D projection by pixel membership.

Note: $\log2(25) = 2 \log2(5)$ is not an integer. Therefore *5 bits* are needed to address one of *25 pixels* (integer ceiling: $\mathrm{ceil}(4.6439) = 5$ *bits*), while *4 bits* can address only *16*. Five bits over-specify by $2^5/25 = 32/25 = 1.28x$ — each pixel uses *4/5* of the address space available in *5 bits*, with *7 codes* unused.

VII.2 The 25 Pixels as a Structured Address Space

The *25 pixels* in the Source x Boundary projection are not all informationally equivalent. They fall into four categories based on whether each axis coordinate is in the clearance margin or not:

Derivation V.25PA · **The Four Pixel Categories**

Category 1 — Both axes in primitive territory *(x_S in [0,E], x_B in [0,F])*:

```
Source primitive: 8 of 10 steps, Boundary primitive:
6 of 10 steps                                              (2)
```

```
Category 1 pixels: (8/10 / δ) x (6/10 / σ) = 8 x (6/4) =
not integer                                                (3)
```

A Structural Pixel spans δ (Source) x σ (Boundary). The number of whole pixels fitting in the Source-primitive (8 steps) x Boundary-primitive (6 steps) region: 8 steps x 6 steps / (1 step x 4 steps) = 12 pixels.

Category 2 — Source in clearance, Boundary in primitive *(x_S in [E,1], x_B in [0,F])*:

```
Source clearance: 2 steps, Boundary primitive: 6 steps     (4)
```

```
Category 2 pixels: 2 x 6/4 = 3 pixels
(the Source-margin x Boundary-primitive region)            (5)
```

Category 3 — Source in primitive, Boundary in clearance *(x_S in [0,E], x_B in [F,1])*:

```
Source primitive: 8 steps, Boundary clearance: 4 steps     (6)
```

```
Category 3 pixels: 8 x 4/4 = 8 pixels
(the Source-primitive x Boundary-margin region)            (7)
```

Category 4 — Both in clearance *(x_S in [E,1], x_B in [F,1])* — the Ontological Shadow region:

```
Source clearance: 2 steps, Boundary clearance: 4 steps        (8)
Category 4 pixels: 2 x 4/4 = 2 pixels
(the Ontological Shadow pixels)                               (9)

 Total: 12 + 3 + 8 + 2 = 25 pixels  ok                        (10)
```

VII.3 The Ontological Shadow as Minimum-Probability 2D Cell

The Ontological Shadow *OS = [E, E+δ] x [F, 1] x [0,1]* (D.OS, Essay II) spans exactly one Source-margin step (from *x_S = E to x_S = E + δ*) and the full Boundary margin (from *x_B = F to x_B = 1*). In the 2D Source x Boundary projection, the *OS* occupies a region of area *δ x σ = P = 1/25*.

The *OS* occupies exactly one Structural Pixel in the Source x Boundary projection — specifically the single pixel at the lower boundary of the Source clearance combined with the full Boundary clearance. However, comparing it to the other 24 pixels, the *OS* pixel is special: it is the pixel that lies simultaneously at the entry to the Source clearance AND in the full Boundary clearance.

Theorem T.OSI · **The Ontological Shadow as Minimum-Probability 2D Cell**

In the 25-pixel tiling of the Source x Boundary projection:

Category 4 (*OS* region) contains 2 pixels: the two Source-clearance steps each spanning the full Boundary clearance. These are the most 'specific' cells: they lie simultaneously in the Source clearance AND the Boundary clearance.

$$P(\text{random position in Category 4}) = (I\_min \times \sigma) = (1/5 \times 2/5) = 2/25 = 0.08 \quad (11)$$

$$P(\text{random position in Ontological Shadow specifically}) = P = 1/25 = 0.04 \quad (12)$$

The *OS* pixel has probability *1/25* in the 2D projection — the same as every other Structural Pixel (since all pixels have equal area *P*). Its distinctiveness is not its probability but its position: it is the one pixel that initiates Source clearance (first Source-margin step) while being fully inside Boundary clearance.

Informational interpretation: the *OS* pixel carries the same pixel-address information as any other pixel — *log2(25) ≈ 4.64 bits*. But its structural position makes it informationally significant: it is the minimum geometric expression of nothing's dual clearance requirement. It is the cell where BOTH the Source face AND the Boundary face are simultaneously in their clearance territory — the geometric location where nothing's internal structure is at maximum structural tension.

VII.4 The Joint Clearance Information

The information of being in the joint Source-AND-Boundary clearance (Category 4 in the pixel taxonomy) can be computed from the probability *2/25 = 0.08*:

Derivation V.JCI · **Joint Clearance Information**

Category 4 (both Source and Boundary clearance) occupies 2 of 25 pixels.

$$P(\text{Category } 4) = 2/25 = 0.08 \tag{13}$$

$$I(\text{Category } 4) = -\log_2(2/25) = \log_2(25/2) = \log_2(12.5) = 3.6439 \text{ bits} \tag{14}$$

For the Ontological Shadow specifically (first Source-margin step x full Boundary margin):

$$P(OS) = 1/25 = P = 0.04 \tag{15}$$

$$I(OS) = -\log_2(1/25) = \log_2(25) = 4.6439 \text{ bits} \tag{16}$$

Verification via axis independence (since Source and Boundary axes are independent, T.GA):

$$I(OS) = I(\text{first Source-margin step}) + I(\text{full Boundary margin}) \tag{17}$$

$$= log2(n/1) + log2(n/\sigma\_steps) \tag{18}$$

$$= log2(10) + log2(10/4) \tag{19}$$

$$= 3.3219 + 1.3219 = 4.6439 \text{ bits ok} \tag{20}$$

VII.5 The Structural Pixel Count and the Clearance Architecture

The count $1/P = 25$ *pixels* has a structural derivation: $25 = (1/\delta) \times (1/\sigma) = n \times (1/c\_steps) \times (n/n) = n / c\_steps \times n = 10/4 \times 10 = 25$. But $10/4$ is not an integer — so the 25-tiling should be verified more carefully:

Derivation V.PC · **The 25-Pixel Count: Structural Derivation**

The unit square has area 1. Each Structural Pixel has area $P = \delta \times \sigma = (1/n) \times (c\_steps/n) = c\_steps/n\text{^}2 = 4/100 = 1/25$.

$$1 / P = n\text{^}2 / c\_steps = 100 / 4 = 25 \tag{21}$$

This is exact and integer because $c\_steps = 4$ *divides* $n\text{^}2 = 100$ exactly. The divisibility $4 \mid 100 = 4 \times 25$ is a structural property: $c\_steps = 4 = 2\text{^}2$, *and* $n\text{^}2 = 10\text{^}2 = 4 \times 25 = 2\text{^}2 \times 5\text{^}2$. Since 4 | 100, the *25 pixels* tile the unit square exactly with no partial pixels.

Structural check: $a\_steps = 2$, $c\_steps = 4$.
Does a_steps divide n^2? $n\text{^}2/a\_steps = 100/2 = 50$. (22)

A Source-based pixel $(\delta \times I\_min)$ would tile the unit square into 50 *pixels*. (23)

```
The Structural Pixel (Boundary-based) gives 25 tiles;
the Source pixel gives 50 tiles.                                    (24)

Ratio: 50/25 = 2 = c_steps/a_steps = 4/2 = c/a
(the ratio from T.UC)                                               (25)
```

Part VIII — The Grand Informational Partition

Parts I through VII have computed information quantities for every structural region of nothing's lattice. Part VIII assembles these into the Grand Informational Partition: the complete information-theoretic characterisation of all positions in *NS*, organised by their structural category.

VIII.1 The Complete Information Register of *NS*

Every position in *NS* can be characterised by three independent binary questions — one per axis — and a set of structural-region membership questions. The independent binary questions are the three clearance-labels: Is x_S in the Source clearance? Is x_B in the Boundary clearance? Is x_R in the Remainder clearance? These three bits (with entropy $H_S + H_B + H_R = 2.572$ *bits*) are the fundamental informational structure of any position in *NS*.

Theorem T.GIP · **The Grand Informational Partition**

The complete information register of a position (x_S, x_B, x_R) in *NS*:

Level 1 — Full address:

```
3 x log2(10) = 9.97 bits
(full coordinate, specifying one of 1000 positions)
```
(1)

Level 2 — Three clearance-labels (independent binary bits, one per axis):

```
H_S + H_B + H_R  =  0.7219 + 0.9710 + 0.8793  =
2.5722 bits
```
(2)

Level 3 — Pixel address in Source x Boundary projection:

```
log2(25) = 4.6439 bits
(one of 25 Structural Pixels)
```
(3)

Level 4 — TSV membership:

```
I(in TSV) = log2(125/42) = 1.5735 bits
(TSV occupies 336/1000 positions)
```
(4)

```
I(not in TSV) = log2(125/83) = 0.5867 bits
(CC occupies 664/1000 positions)
```
(5)

Level 5 — Clearance margins (per-axis):

```
I(in Source clearance) = log2(10/2) = log2(5) =
2.3219 bits
```
(6)

```
I(in Boundary clearance) = log2(10/4) = log2(2.5) =
1.3219 bits
```
(7)

```
I(in Remainder clearance) = log2(10/2.984) =
log2(3.352) = 1.7448 bits
```
(8)

Level 6 — Ontological Shadow:

```
I(in OS) = log2(25) = 4.6439 bits
(most informative 2D clearance cell)
```
(9)

Level 7 — Forbidden Ceiling (Informational Bound):

```
log2(3) = 1.5850 bits
(forbidden by T.NE — degenerate face-label ceiling)
```
(10)

VIII.2 The Three Information Orderings

Essay III has derived three orderings that all agree:

Principle P.3IO · **The Three Consistent Orderings**

Clearance ordering (from Essay I, P.HI):

$$\sigma \ > \ \Lambda \ > \ I\_min \quad (0.400 > 0.2984 > 0.200) \tag{11}$$

Axis-entropy ordering (from Essay III, T.FE):

$$H\_B \ > \ H\_R \ > \ H\_S \quad (0.9710 > 0.8793 > 0.7219 \text{ bits}) \tag{12}$$

Clearance bit-address ordering (from Essay III, P.CBA):

$$\text{Boundary bits} \ > \ \text{Remainder bits} \ > \ \text{Source bits } (2.000 > 1.577 > 1.000 \text{ bits}) \tag{13}$$

All three orderings are consistent. The structural hierarchy of Essay I (largest clearance at Boundary, smallest at Source) is expressed informationally in three equivalent ways: as absolute clearance sizes, as axis partition entropies, and as clearance margin addressing costs. The informational register does not add new hierarchy — it re-reads the existing hierarchy through counting.

VIII.3 The Three Key Informational Ratios

Derivation V.KIR · **Three Key Informational Ratios**

Ratio 1 — Boundary/Source bit-address ratio:

$$\text{Boundary bits / Source bits} = \log 2(4) / \log 2(2) = 2/1 = 2 \tag{14}$$

$$= c\_steps / a\_steps = 4/2 = 2 \quad \text{(from T.UC, Essay I)} \tag{15}$$

Ratio 2 — TSV information relative to Forbidden Ceiling:

$$\log 2(3) - I(TSV) = \log 2(3) - \log 2(125/42) = \log 2(126/125) = 0.0115 \text{ bits} \tag{16}$$

$$= \text{the informational gap between actual and degenerate } TSV \text{ occupation} \tag{17}$$

Ratio 3 — Dimensional compression ratio:

$$I(\text{preserved}) / I(\text{full}) = 1/d = 1/3 \quad \text{(single-axis information as fraction of full)} \tag{18}$$

$$= \beta\_c = 1/d = 1/3 \quad \text{(same as the continuum exponent)} \tag{19}$$

VIII.4 The Informational Equivalents of the Three Failure Modes

Theorem T.IFMD · **The Three Failure Modes in the Informational Register**

Deterministic Crystal ($\sigma = 0, F = 1$):

```
H_B -> 0 bits:
Boundary axis carries no clearance information                (20)

I(Boundary clearance) -> log2(10/0) =
undefined (0 clearance positions)                             (21)

The Boundary axis is informationally null:
no position is in clearance                                   (22)
```

Total Dissolution (σ -> 1, F -> 0):

```
H_B -> 0 bits:
Boundary axis again carries no clearance information          (23)

I(Boundary clearance) -> log2(10/10) =
0 bits (all positions in clearance)                           (24)

The Boundary axis is informationally null:
every position is in clearance — no distinction possible      (25)
```

Causal Closure Seizure ($\Lambda = 0, \delta = 1$):

```
H_R -> 0 bits: Remainder axis carries no clearance
information                                                   (26)

The dimensional collapse fills the entire Remainder axis:
no residual information remains                               (27)
```

Structural observation: all three failure modes correspond to an axis-entropy approaching 0. $H = 0$ means complete predictability — no information in the clearance-label because the outcome is certain. The failure modes are the states of maximum informational certainty in the structurally worst sense: an axis that is certain to be entirely in one territory or the other. Nothing's viable structure requires all three axis-entropies to be strictly positive.

Conclusion

What Has Been Derived

Essay III has derived the complete informational architecture of nothing's structural space from the discrete lattice of Essays I and II. Every information quantity is a counting result: position-counts followed by $log2$. No external theory was imported.

Part I: Bit-addressing of NS: $n = 10$ requires $log2(10) \approx 3.32$ *bits* per axis; full NS requires $\lceil 3\ log2(10) \rceil = 10$ *bits*. Three faces require between 1 and 2 *bits* to address (T.FA), with the maximum $log2(3) \approx 1.585$ *bits* achievable only in the forbidden degenerate state.

Part II: The face-entropy: each axis has a binary-partition entropy $H(p, 1-p)$ where p is the clearance fraction. $H_B = 0.9710$ *bits*, $H_R = 0.8793$ *bits*, $H_S = 0.7219$ *bits*. The Forbidden Ceiling $log2(3)$ is the entropy of the degenerate equal-face partition, forbidden by T.NE. T.FE establishes that actual axis-entropies are below this ceiling.

Part III: Clearance addressing: $Source = log2(2) = 1$ *bit* exactly (minimum binary distinction, T.MBD); $Boundary = log2(4) = 2$ *bits* exactly (T.BC2); $Remainder = log2(2.984) = 1.577$ *bits* (non-integer, sub-lattice). The one-bit step between Source and Boundary (T.BOS) is the informational expression of $c/a = 2$ (T.UC).

Part IV: Axis-entropy hierarchy T.FE confirmed: $H_B > H_R > H_S$ matches clearance ordering exactly. $H_B > H_{degenerate}$ confirms the Boundary axis is informationally richer than the degenerate threshold (T.BD). Joint entropy $H_{joint} = 2.572$ *bits* $= 85.7\%$ of theoretical maximum 3 *bits*.

Part V: TSV information: $I(TSV) = log2(125/42) = 1.5735$ *bits* — 0.0115 *bits* below the Forbidden Ceiling $log2(3)$. The deficit is $log2(126/125)$, arising from TI exceeding $1/3$ by $1/375$. A structural bonus: $TI - (2/3)^3 = 0.336 - 0.296 = 0.040 = P = Structural\ Pixel$.

Part VI: Dimensional collapse as information compression: preserves exactly *1/d = 1/3* of full coordinate information (T.DCI). This fraction is the continuum exponent *β_c = 1/d* — the same structural constant from two different registers simultaneously.

Part VII: Structural Pixel information: *log2(25) = 4.64 bits* per pixel (T.SPI). The *25 pixels* partition into four categories by clearance-membership. The Ontological Shadow is the minimum-probability 2D clearance cell, occupying *1* of *25 pixels*. The Source-based pixel count (50) is exactly double the Boundary-based pixel count (*25*), consistent with *c/a = 2*.

Part VIII: Grand Informational Partition: all information quantities assembled. Three consistent orderings (clearance, axis-entropy, bit-address) all agree. Three failure modes re-expressed as axis-entropy boundaries (all drive H to *0* on the relevant axis).

The Terminal Result

The informational register adds nothing new to nothing's architecture. It translates what is already fully derived — in logical, algebraic, and geometric registers — into the language of counting. The same structural results appear as bit-counts, *log2* values, and entropy calculations. The consistency is not surprising: every number in the informational register is derived from counting positions on the discrete lattice of Essay II, which was itself derived from the logical architecture of Essay I.

What the informational register contributes is precision: it makes explicit what was implicit. The Forbidden Ceiling *log2(3)* makes explicit that T.NE is an informational constraint — nothing's three faces must NOT achieve maximum informational uniformity. The one-bit step between Source and Boundary makes explicit that *c/a = 2* is a binary step — the Boundary is informationally richer by exactly one binary distinction. The *1/d = 1/3* compression ratio makes explicit that the continuum exponent and the information-preservation fraction are the same structural constant.

Theorem T.ITA · **The Informational Terminal Architecture**

Nothing's informational architecture, completely derived from lattice counting:

Full address:
10 bits (*1000* positions in *NS*) (28)

Forbidden ceiling:
log2(3) = 1.585 bits (degenerate face-label, forbidden by T.NE) (29)

Axis entropies:
H_B = 0.971, H_R = 0.879, H_S = 0.722 bits (30)

Joint entropy:
2.572 bits (= 85.7% of maximum 3 bits) (31)

Source clearance:
1 bit (minimum binary distinction, 2 positions) (32)

Boundary clearance:
2 bits (one step richer than Source, *c/a = 2*) (33)

Remainder clearance:
1.577 bits (non-integer, sub-lattice, from collapse) (34)

TSV information:
1.574 bits (below Forbidden Ceiling by *log2(126/125)*) (35)

Compression ratio:
1/d = 1/3 (fraction preserved by dimensional collapse) (36)

Pixel address:
log2(25) = 4.644 bits (one of *25 Structural Pixels*) (37)

Every value is derived from counting on the $n = 10$ discrete lattice. Zero free parameters. Zero external information-theoretic framework. The informational ground is nothing's structure read through counting alone.

Bibliography

The following works are cited for structural isomorphism confirmation only. None serve as derivational foundations for the present framework. All results herein are derived from the internal logic of the Triad and the normalised field alone.

Internal series references

Pretorius, E. (2026). *Gradientology - Foundations of the Primordial Triad*: Treatises I-XVI. Zenodo.

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ADDENDUM: UNIVERSAL TERMS OF ENGAGEMENT FOR THE RESIDUAL SUITE

Preamble This addendum serves as the binding and immutable interpretive constraint specifically for the **Residual Suite** (Validation Protocols and Ledger Audits for the 20% Minimum Bit-Budget, 30% Ontological Gap, and 40% Structural Volatility). Its purpose is to formalize the absolute universality of these geometric clearances, explicitly foreclosing any attempt to restrict them to a specific domain, timeframe, or material.

CONSTRAINT 1: Phase Invariance (The Permanent Mechanics)

The three residuals ($I\_min \approx 0.2, \Lambda \approx 0.298, \sigma = 0.4$) are not pre-kinetic artifacts or temporary conditions of the primordial stasis. They are the permanent, load-bearing mechanical clearances of the universe. Analysts are strictly prohibited from treating these gaps as confined to Phase I. They operate continuously across all phases—preventing the Deterministic Crystal in Phase II and acting as the Velocity/Saturation blocks that mandate perpetual novelty in Phase III. The gaps are the permanent operating tolerance of the engine.

CONSTRAINT 2: Scale Invariance (The Scalar Mandate)

The structural clearances apply with absolute isomorphic fidelity at every scalar resolution of the **Veldt.** Readers and analysts are prohibited from treating the residuals as exclusively "quantum" phenomena or exclusively "cosmological" phenomena. The 40% Structural Volatility is identically the source of Heisenberg's Uncertainty Principle at the micro-scale, the mandated tolerance in macroscopic precision engineering, and the limit of resolution in computational mathematics. The math is scale-invariant.

CONSTRAINT 3: Substrate Neutrality (The Purge of Material Bias)

The residuals are pure mathematical inevitabilities of a discrete lattice. They must never be reified by attributing them to a specific material substrate. The 20%, 30%, and 40% clearances govern biological ecosystems, artificial intelligence networks, silicon processing, and the astrophysical void equally. Importing material, biological, or anthropic properties to explain the existence of these gaps is a prohibited circularity.

CONSTRAINT 4: Epistemic Foreclosure (Anti-Reification of Uncertainty)

Regardless of the phase, scale, or substrate being analyzed, no reader may interpret these three residuals as measurement errors, statistical noise, or "epistemic limits" arising from observer ignorance. They are derived **Mechanical Necessities**. To claim these gaps can be "closed" by better measurement, or to treat them as probabilistic noise, is to fundamentally misunderstand the engine. **The residuals are not what the engine lacks — they are what the engine is.**

CONSTRAINT 5: Final Demarcation and Logical Necrosis

These Terms of Engagement are non-negotiable. Any attempt to localize these clearances to a specific phase, restrict them to a specific scale, tie them to a specific substrate, or dismiss them as probabilistic contingencies constitutes a formal epistemic breach classified as **Logical Necrosis**.